CAPSTONE PROJECT PRESENTATION

URBAN CAB FARE ANALYSIS

The Urban Cab Fare dataset contains cab booking details of 1,000 users across six major cities in India, covering the period from August 2025 to August 2026

PRESENTED BY
Sanal K P

TIMELINE
Oct 2026

PRESENTATION OVERVIEW

Executive Roadmap

01

Scope & Objective

Business context, urban mobility setup, and core objectives.

02

Dataset & Prep

Data cleaning, feature engineering, and pipeline architecture.

03

Demand Analysis

Revenue trends, supply gaps, and city-level ride volume.

04

Insights & Action

Surge patterns, traffic impact, and operational deployment.

MARKET CONTEXT

Urban Mobility Challenges

Demand & Supply Misalignment

Unequal ride requests across cities, days, and peak hours create long passenger wait times or idle driver capacity.

Traffic & Pricing Dynamics

Congestion directly impacts trip duration and surge multipliers. Dynamic fares must balance provider revenue and rider conversion.

1,000

Trip Records Analyzed

- 14 Analytical Features
- Multi-city Coverage
- Granular Hourly Insights

DATA ARCHITECTURE

Dataset Overview & Attributes

Field Category	Key Variables	Analytical Purpose
Trip Info	Distance, City, Vehicle Type	Categorize trip scope and fleet utilization
Pricing Structure	Base Fare, Final Fare, Surge Multiplier	Evaluate yield management & dynamic pricing
Supply & Demand	Ride Requests, Active Drivers	Detect market deficit and driver efficiency
Temporal & Congestion	Day of Week, Hour, Traffic Level	Identify bottleneck hours & high-demand slots

Source Details

Source: Kaggle Dataset (UrbanCabFareData)

Volume: 1,000 Records, 14 Features

Tools: Python (pandas, NumPy, seaborn, matplotlib)

EXECUTION METHODOLOGY

End-to-End Analysis Workflow

1

Data Preparation

Data type validation, missing check, and deduplication.

2

Feature Engineering

Creating Year-Month, Day of Week, and Hour features.

3

Exploratory Analysis

Calculating driver ratios, traffic correlation, and revenue.

4

Visual Strategy

Generating heatmaps, demand curves, and action plans.

FINANCIAL TRACKING

Monthly Revenue Performance

Peak Revenue Month (Apr 2026)

₹41,408.51

Highest Monthly Earnings Achieved

Revenue Variation

Significant seasonal and operational fluctuation across quarters, with March 2026 and October 2025 performing as runner-up peaks.

Tactical Application

Align driver incentive programs and customer promotions precisely around historical revenue expansion windows.

OPERATIONAL CAPACITY

Ride Requests vs. Active Drivers

April 2026 Peak

Ride Requests

10,636

Active Drivers

4,605

Demand-to-Driver Ratio: 2.31x

October 2025 Benchmark

Ride Requests

10,343

Active Drivers

5,119

Demand-to-Driver Ratio: 2.02x

⚠ Core Takeaway

High demand-to-driver ratios signal immediate need for driver onboarding or dynamic shift dispatch to avoid unfulfilled bookings.

GEOGRAPHIC COMPARISON

City Revenue Performance

Top Performing City

Delhi

₹68,961.19

Highest urban market revenue share.

Lowest Yield City

Hyderabad

₹54,069.51

Requires strategic market intervention.

Highest Demand Volume

Pune

19,980

Total ride requests recorded.

TEMPORAL DEMAND

Peak Ride Demand Slots

Critical Day-Hour Heatmap Clusters

Saturday 00:00
Midnight weekend surge

Friday 04:00
Early morning transit

Monday 06:00
Weekday rush start

Saturday 15:00
Weekend afternoon peak

Strategic Action

Deploy targeted shift bonuses during these 4 key time windows to minimize rider drop-off and wait times.

Non-uniform distribution across days.

CONGESTION ANALYSIS

Traffic Impact & Peak Jam Hours



Peak Traffic Hour

9:00 AM

20 Jam-Level Events Recorded

ETA & Service Efficiency

Heavy morning congestion significantly inflates trip duration and impacts driver turnaround time across key corridors.

Dynamic Positioning

Pre-position vehicles outside high-congestion cores prior to 9 AM peak to maintain acceptable ETA promises.

FLEET COMPOSITION

Active Drivers by Vehicle Type

Bike

11,106

Highest Active Drivers

Auto

10,703

High Short-Trip Supply

Hatchback / SUV

Balanced

Mid-Distance Coverage

Sedan

9,484

Lowest Driver Supply

STRATEGIC INSIGHTS

Core Analytical Findings

Market Disparity

Delhi generates maximum overall revenue, whereas Pune leads total request volume, highlighting distinct city monetization profiles.

Capacity Gaps

Demand-supply mismatch peaks during April 2026. Sedan supply remains chronically lagging relative to micro-mobility options.

Temporal Spikes

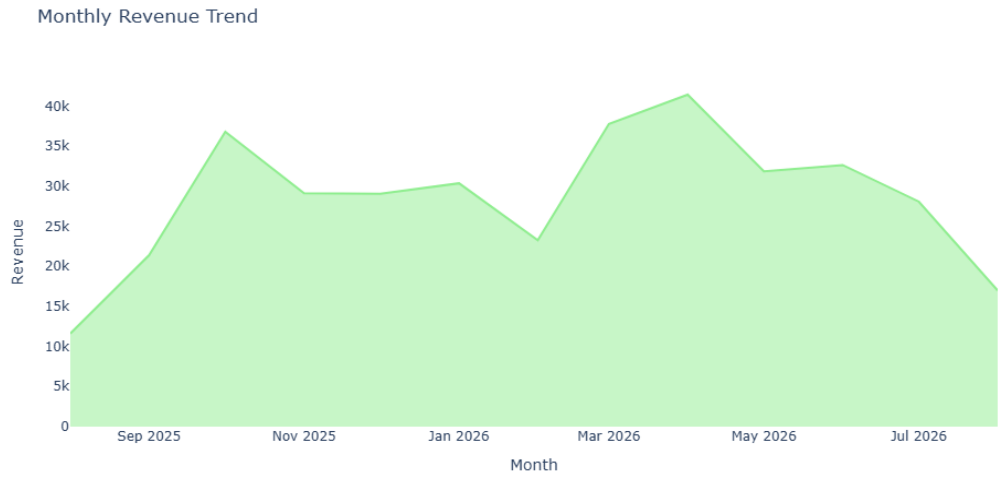
Peak fare yield occurs in off-peak commute hours such as Tuesday 21:00 and Monday 16:00, driven by surge multipliers.

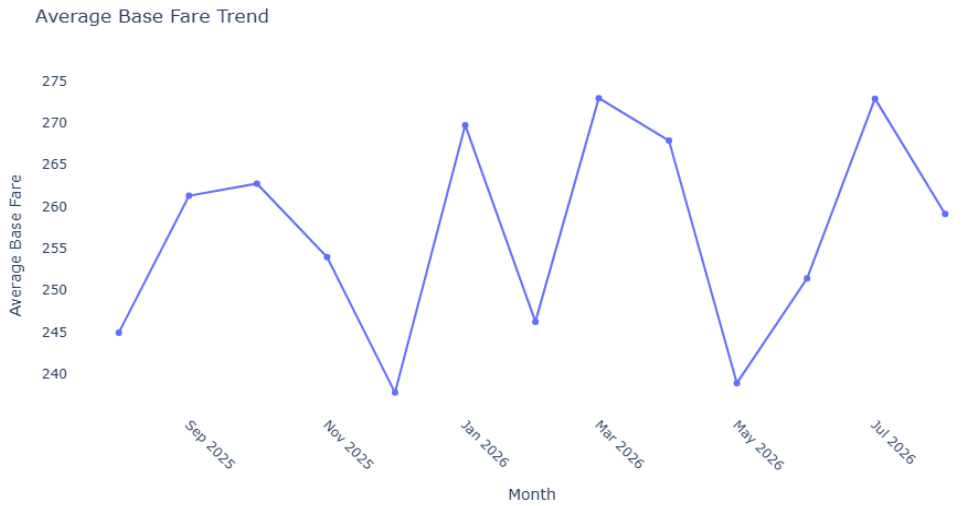
Congestion Friction

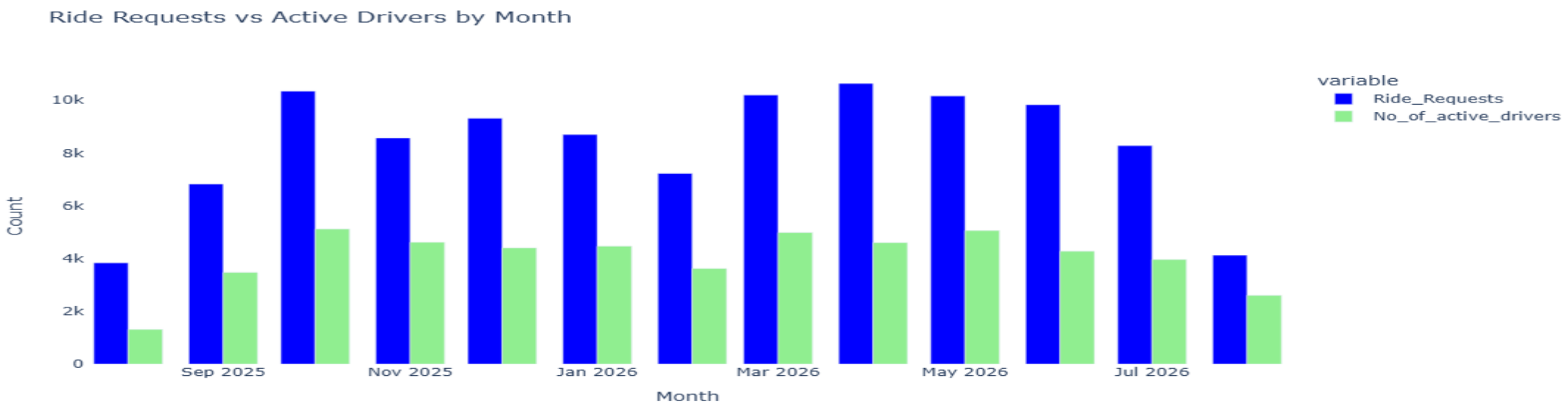
Jam-level traffic concentrates heavily around 9 AM, causing severe dispatch delays if fleet is unmanaged.

Section 1

Monthly Business Performance

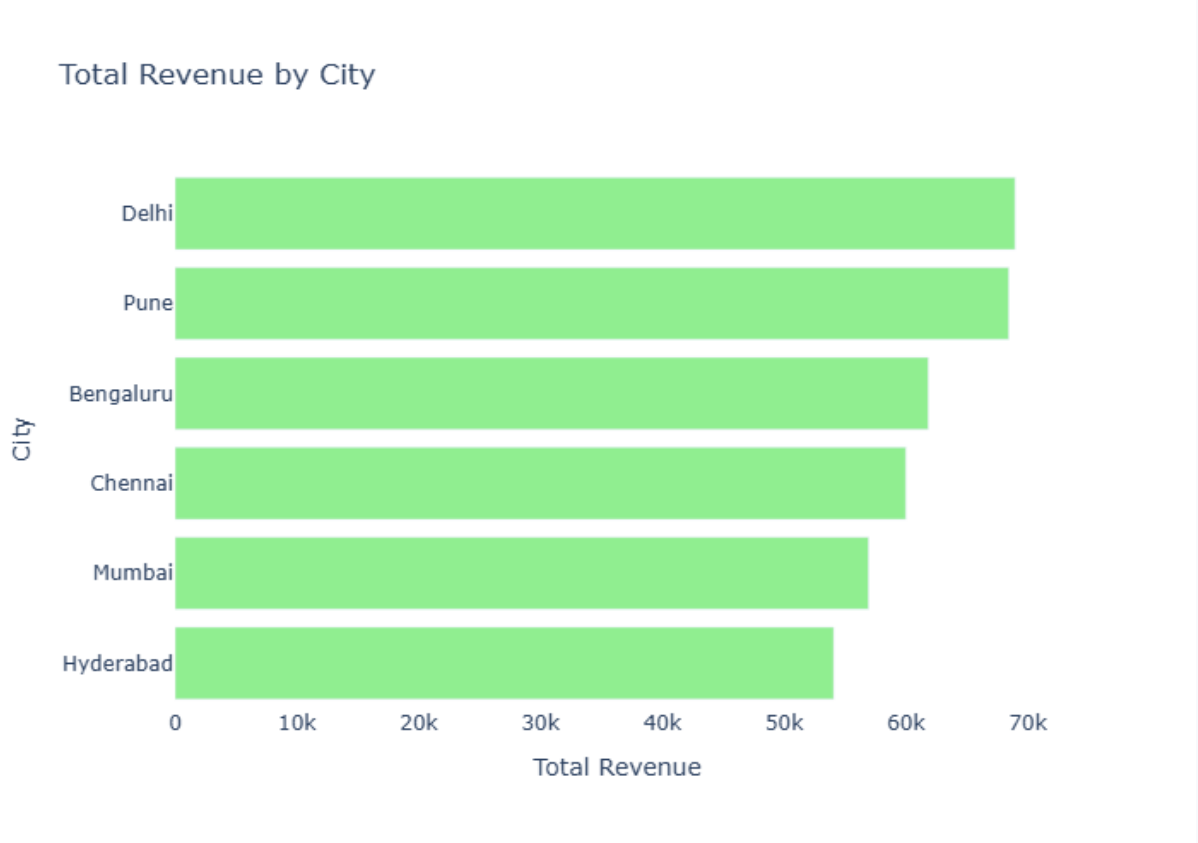


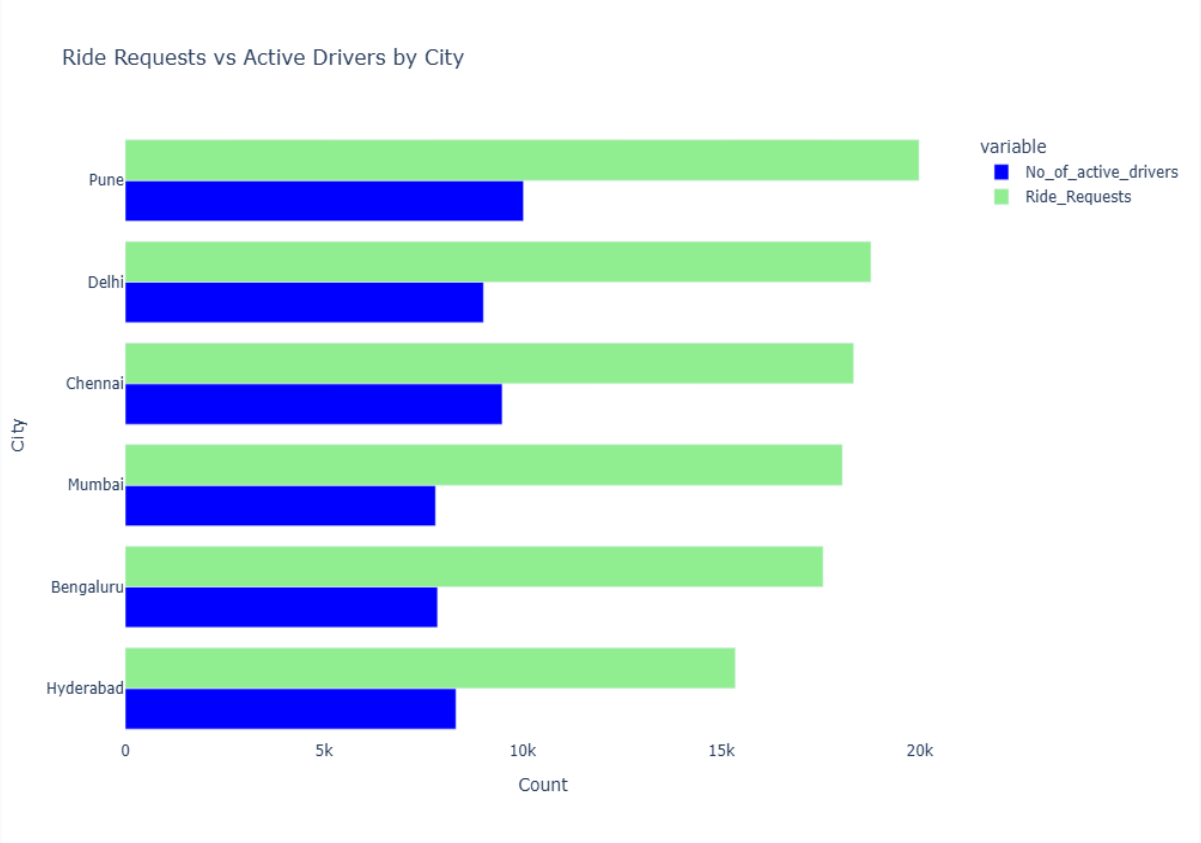




Section 2

City Wise Performance





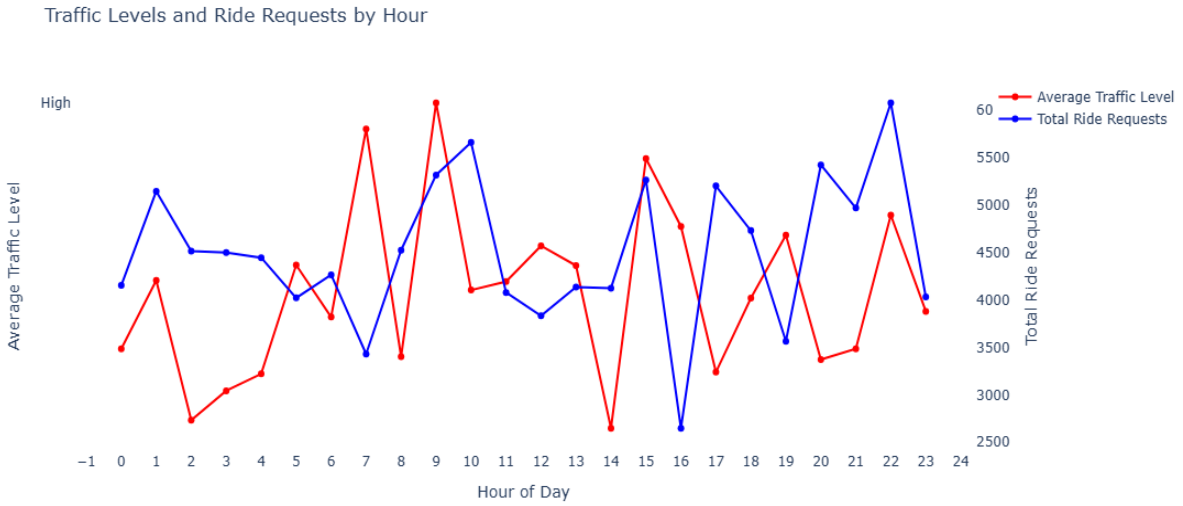
Section 2

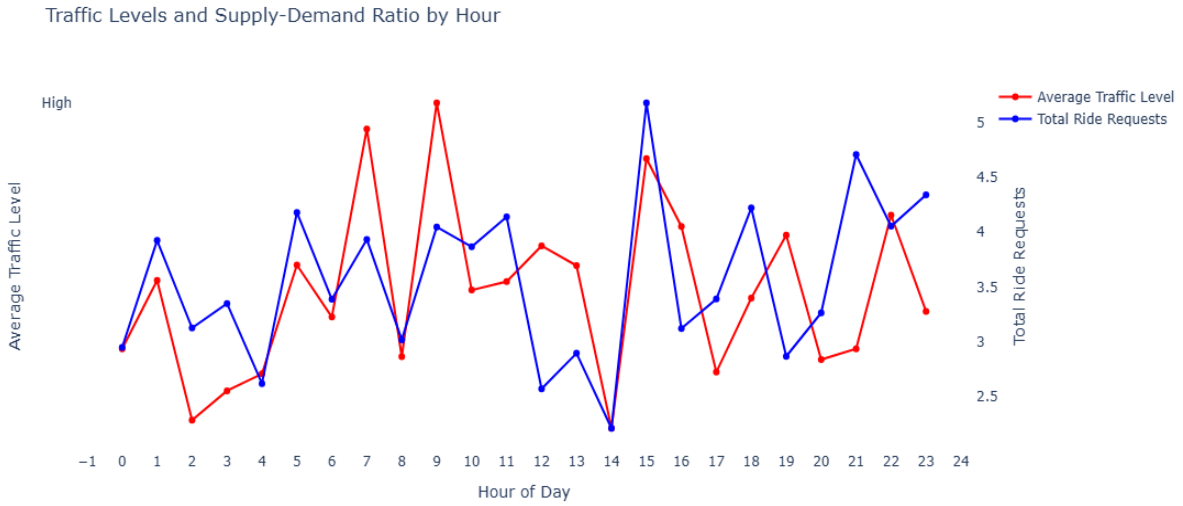
Average Surge multiplier by City

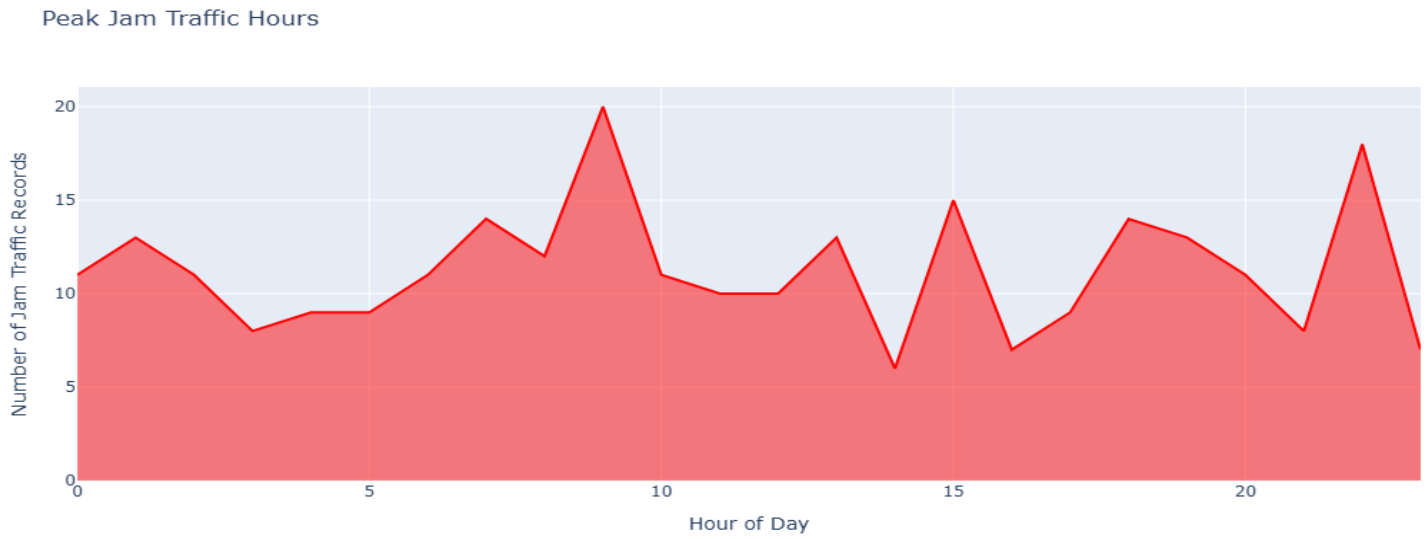


Section 3

Demand, Traffic and Time Patterns

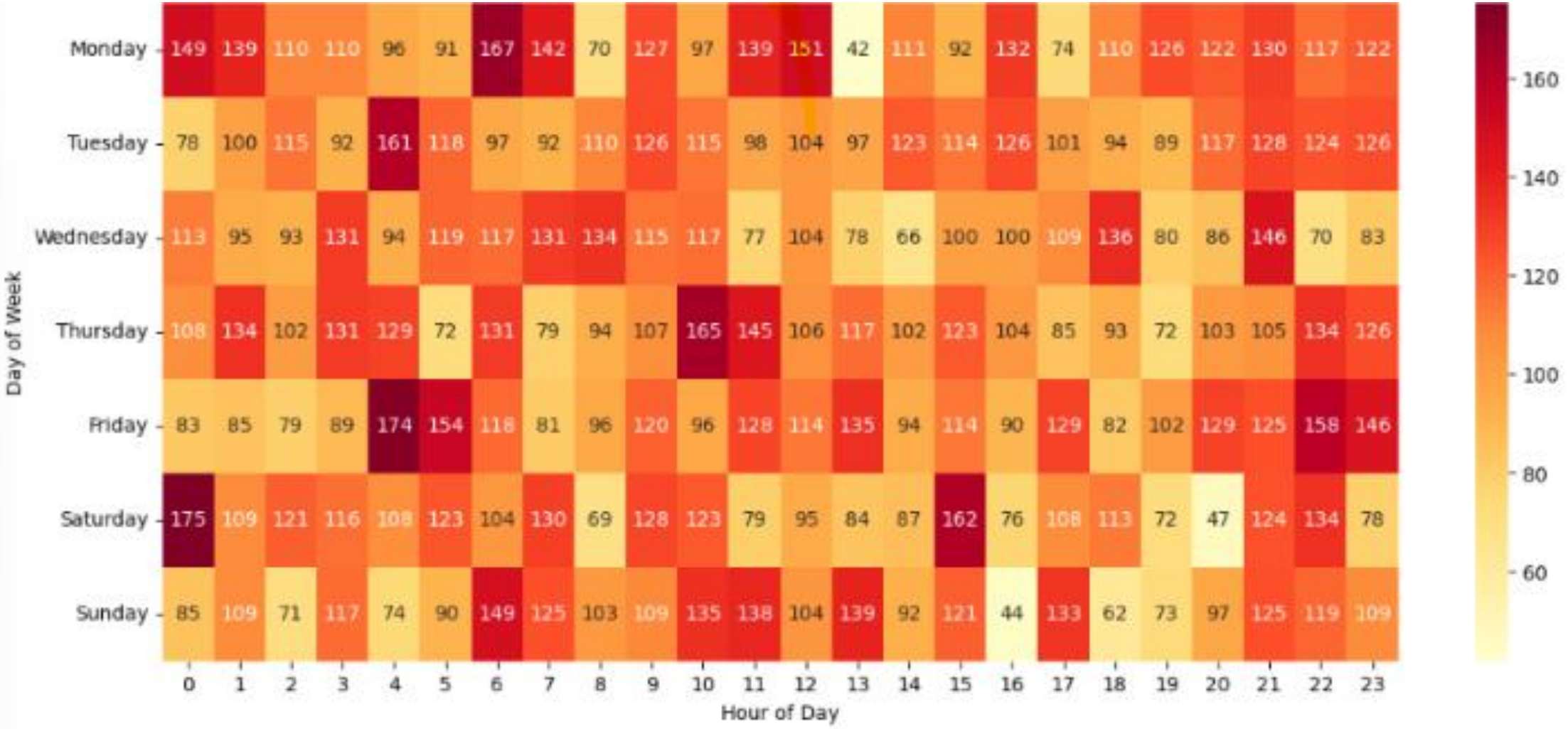






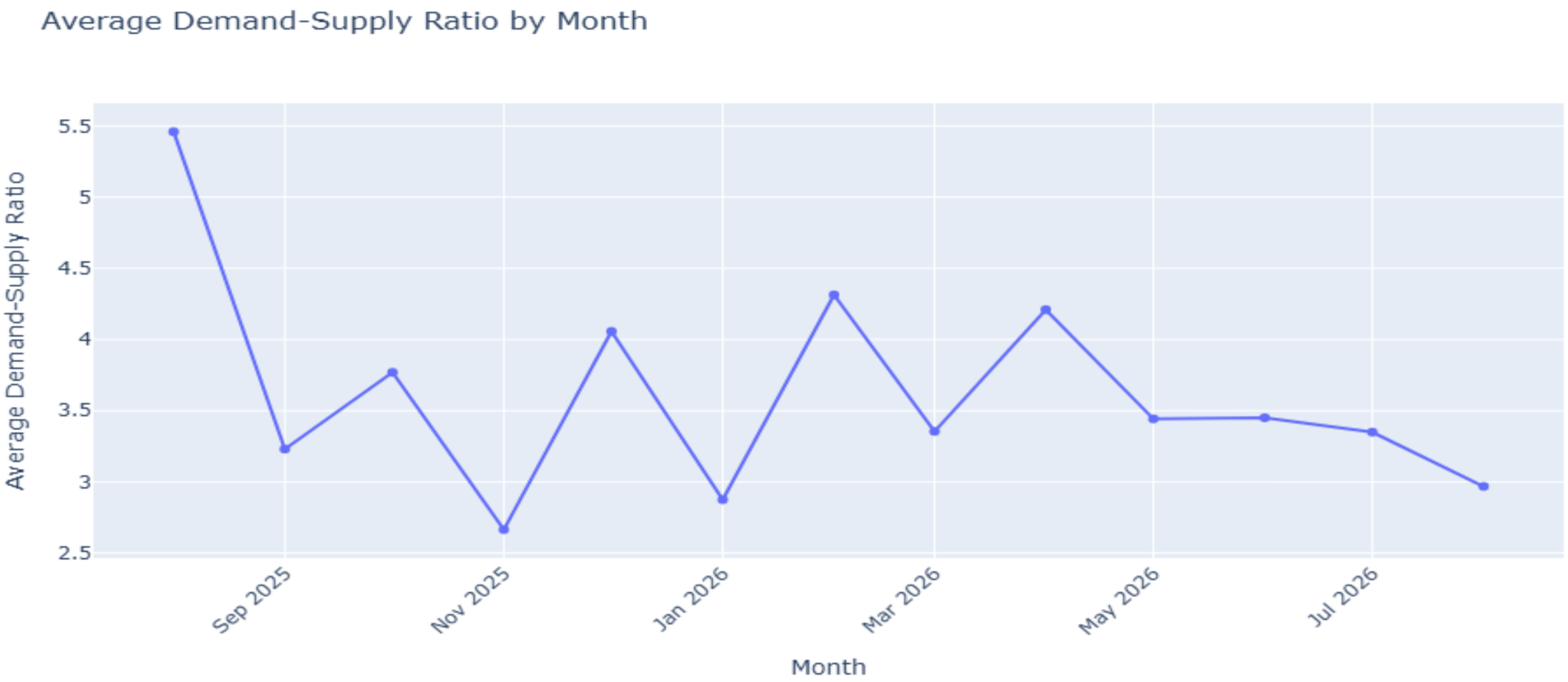
Section 3

Average Ride Requests by Day and Hour



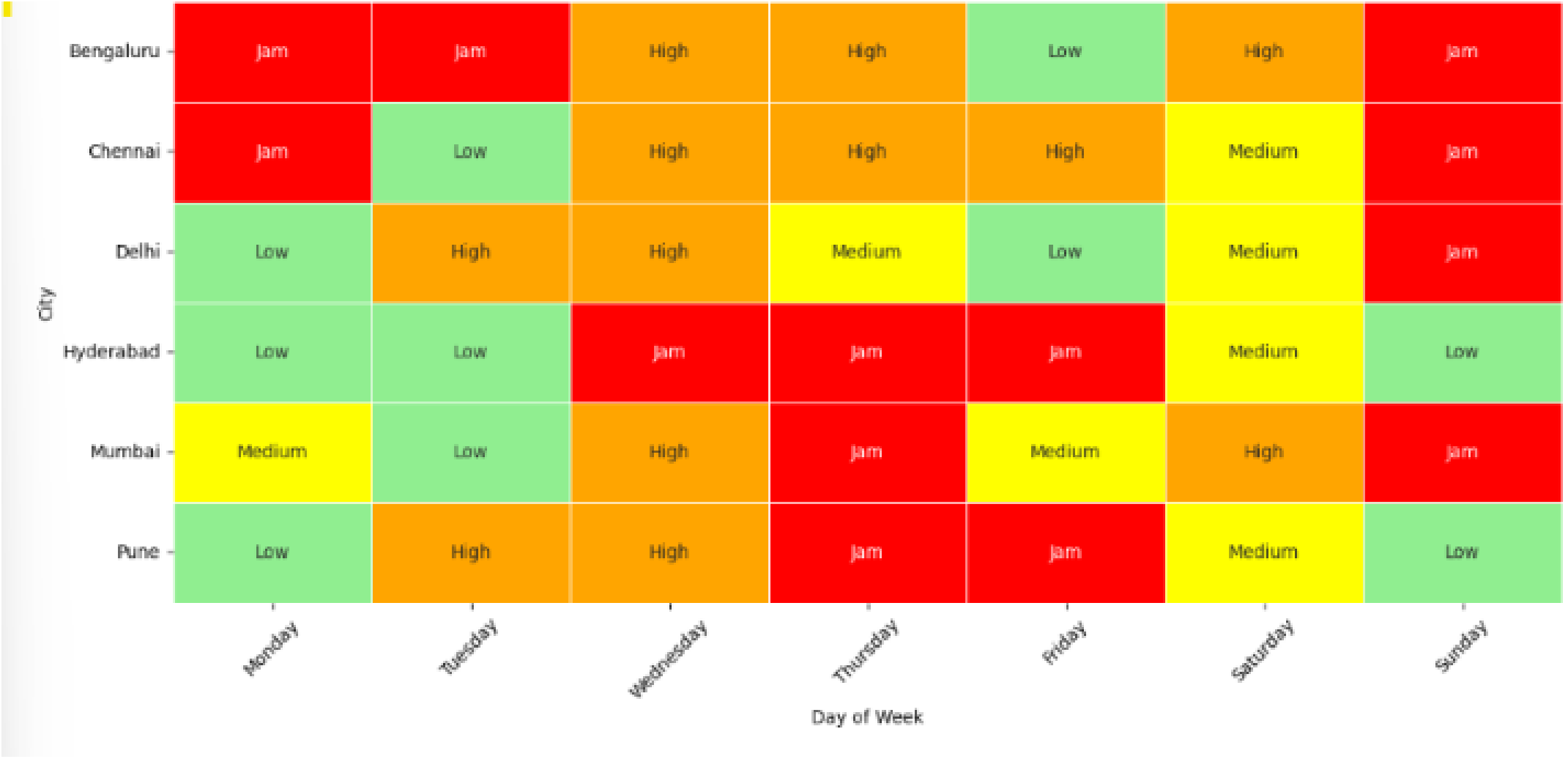
Scection 3

Average demand supply ratio by month



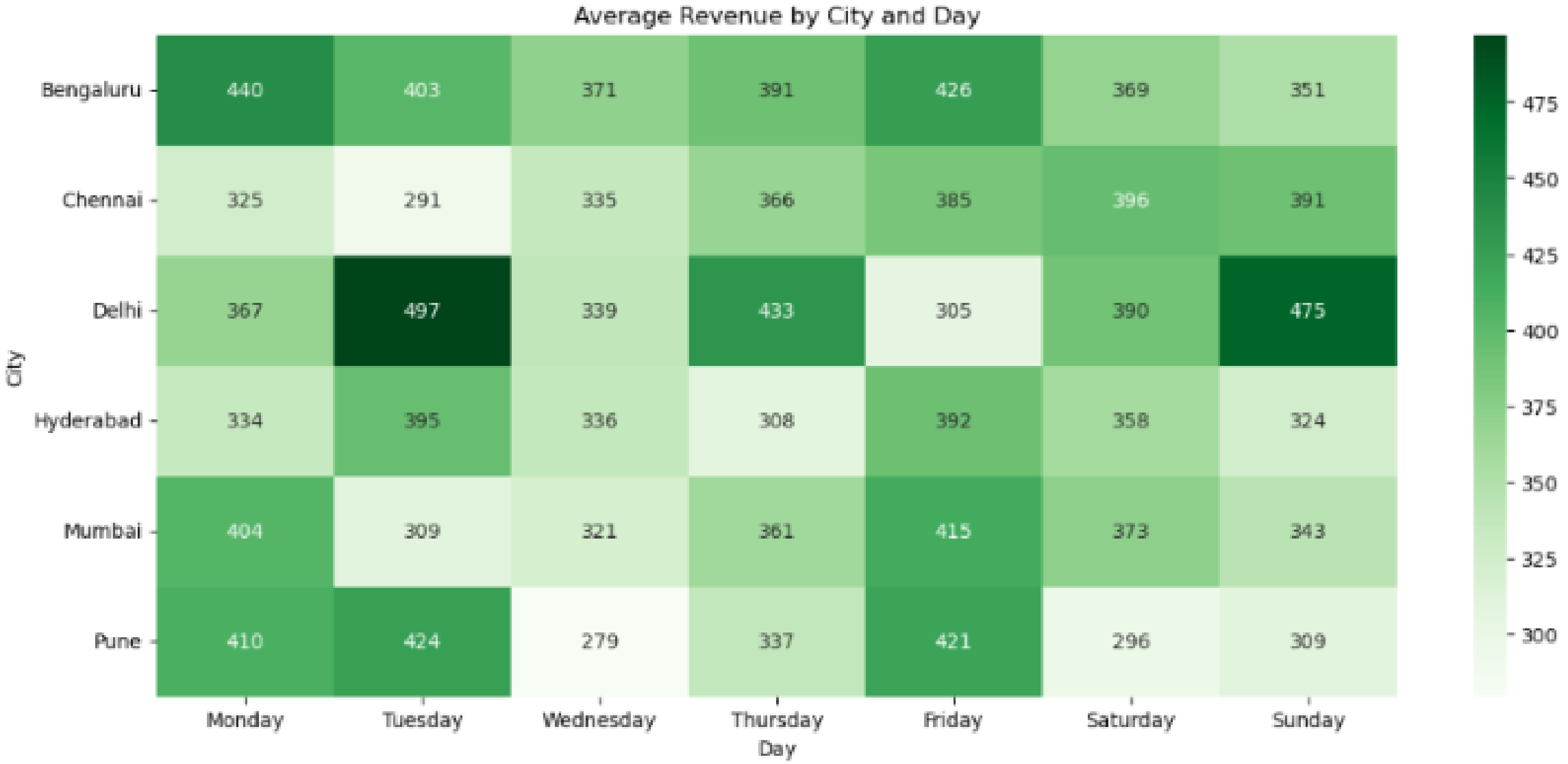
Section 3

Most Common Traffic Level by City and Day



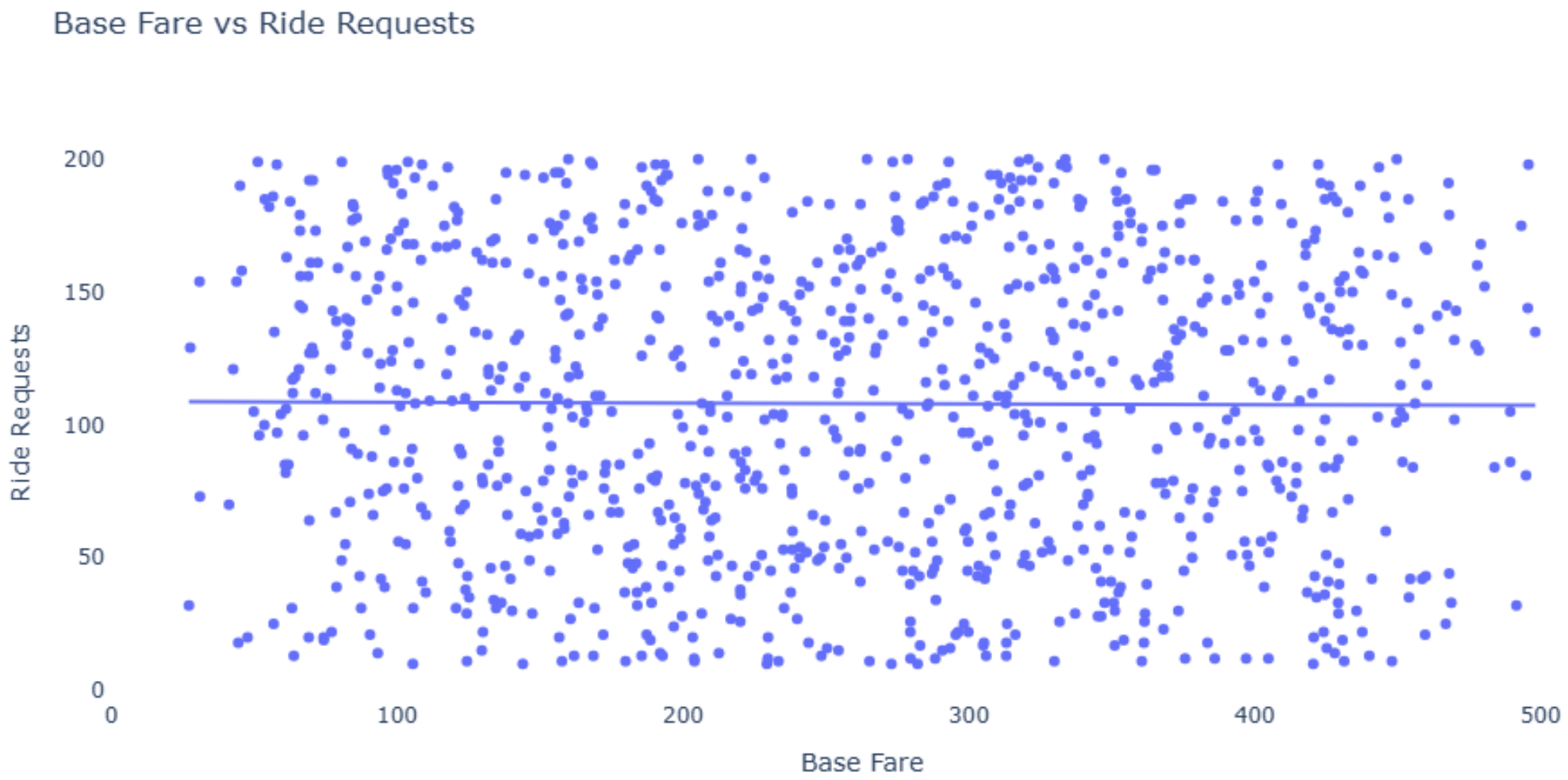
Section 4

Correlations



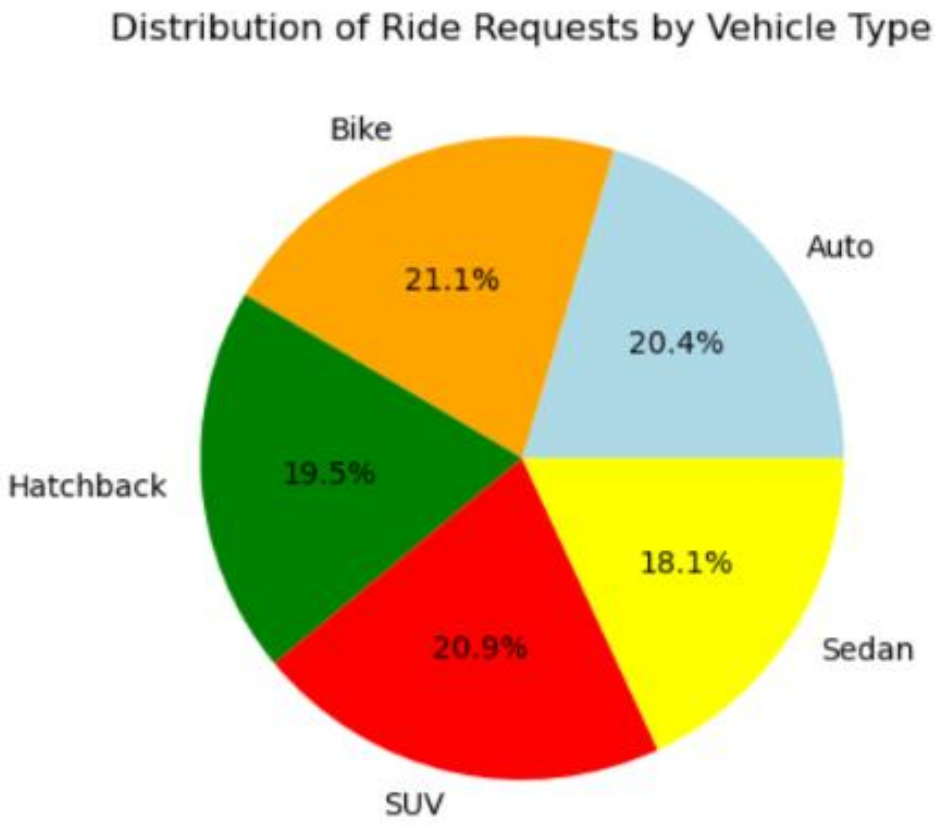
Section 4

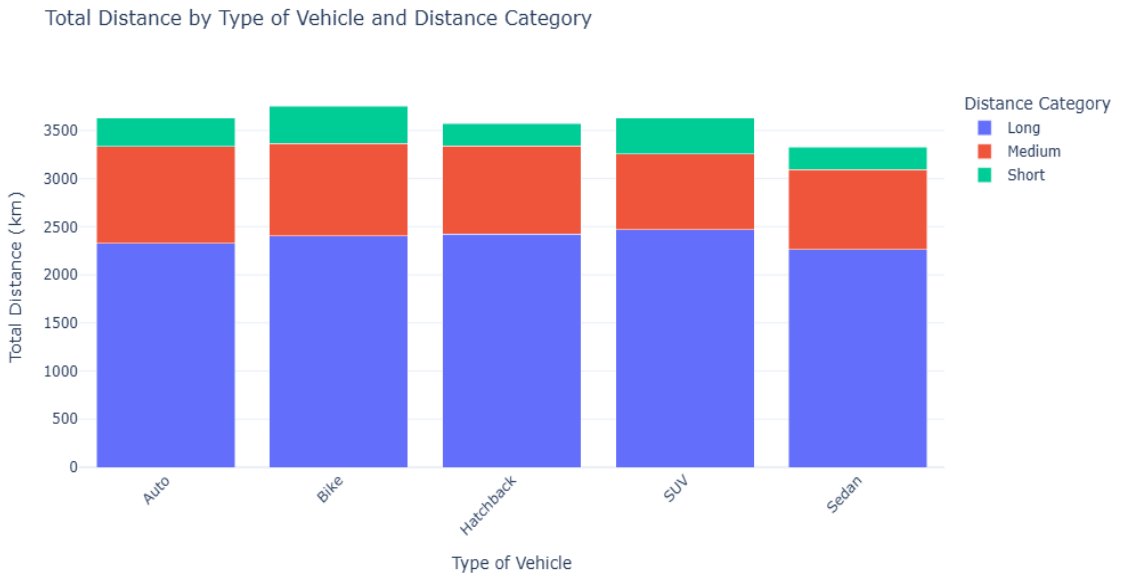
Correlation between base fare and ride requests

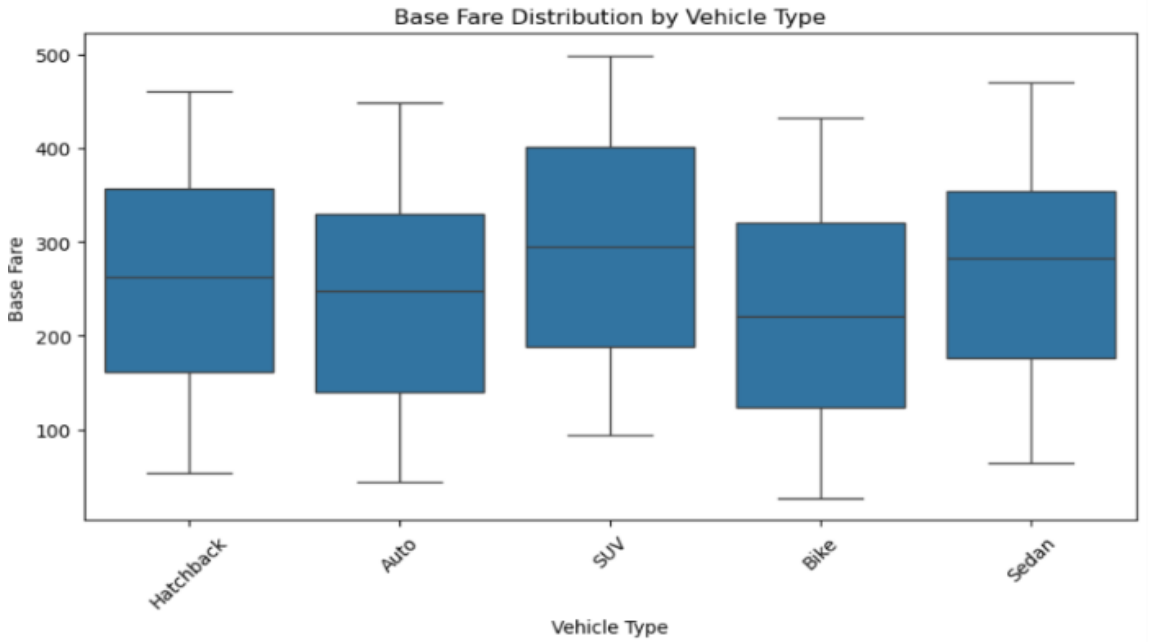


Section 5

Vehicle and Operational Analysis







ACTION PLAN

Strategic Recommendations

Driver Incentives

Incentivize driver deployment during identified peak hours (Sat 00:00, Fri 04:00, Mon 06:00) and scale sedan fleet recruitment.

Dynamic Surge Guardrails

Cap extreme surge during high traffic hours to maintain customer retention while ensuring competitive driver pay.

Real-time Dashboards

Deploy city-level monitoring dashboards tracking traffic, active driver density, and live ride request ratios.

CONCLUSION

Questions & Discussion

Thank you for your time and attention.

Sanal K P • Capstone Presentation 2026